Netflix movies and tv shows_internship.ipynb

Netflix

Importing Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

loading the data

```
data = pd.read_csv("NETFLIX.csv")
data
```

DATA INSPECTION

```
data.head()
```

```
data.tail()
```

```
data.info()
```

```
data.shape
```

```
data.describe()
```

```
data.size
```

Data Cleaning

Finding Missing Values

```
data.isnull().sum()
```

```
data.fillna(0)
```

```
data.dropna()
```

```
data.rename(columns={"show_id":"movie_id"})
data


data.drop_duplicates()
```

Interview question Answers

1. what are missing values and How do you handle them

```
Missing values are data entries that are unavailable,often shown as
NaN ,none or blank cells .I identify them and decide whether to remove
records or fill values using a suitable method such as mean,median or
mode.
    import pandas as pd
    data= pd.read_csv ("data.csv")
    print(data.isnull().sum())
data["Age"= data["Age"].fillna(data["Age"].median())
data["Gender"]=data["Gender"].fillna(data["Gender"].mode()[0])
```

2. How do you treat duplicate records?

```
Duplicate records are repeated entries that can distort analysis. i
check whether they are true duplicates,then use duplicate () to
identify them and drop _duplicates()to remove them.
    print(data.duplicated().sum())
    data=data.drop_duplicates()
# if uniquenees is defined by a column:
data= data.drop_duplicates(suset=["customer_id"])
```

3. Difference between dropna() and fillna() in pandas?

```
dropna() removes rows or column containing missing values ,while
fillna()replaces missing values.dropna()mayreduce the
dataset;fillna()can preserve records when a suitable replacement is
available.
    data=data.dropna()
# remove rows with missing values
data["Age"]= data["Age"].fillna(25)
# replace missing age
```

4 What is outlier treatment and why is it important?

```
Outliers are observations unusually far from the rest of the data.They
can affect the mean ,standard deviation and some models. I identify
them using plots,IQR,or Zscores,then investigate whether they are
errors or genuine values.
    Q1= data["salary"].quantile(0.25)
    Q3=data["salary"].quantile(0.75)
        IQR=Q3-Q1

        lower=Q1-1.5*IQR
        upper=Q3+1.5*IQR

        outliers=data[(data["salary"]< lower),(data["salary"]> upper)]
```

5. Explain the process of standardizing data ?

```
In data cleaning ,standardization means making values an labels
consistent :trimming spaces, using consistent case ,mapping category
varients and making column names uniform.
    data= ["Gender"]=data["Gender"].str.strip().str.lower()
    data= ["Gender"]=data["Gender"].replace
({"m":"male","f":"female"})
   data.columns=(data.columns.str.strip().str.lower().str.replace("
","_"))
```

6. How do you handle inconsistent data formats such as date/time?

```
 I identify differing formats and convert values to a consistent
type.to_datetime()converts dates;invalid values can be coerced to NaT
for review .Numeric strings can be converted with to_numeric().
     data["Date"]=pd.to_datetime(data["Date",dayfirst=True,errors
="coerce")
      data["salary"]=pd.to_numeric(data["salary"],errors="coerce")
      # optional display format:
data["Date"]=data["Date'].dt.strftime("%d-%m-%y")
```

7. What are common data cleaning challenges?

```
Common challenges include missing values,duplicates,inconsistent
formats,incorrect data types,outliers,spelling errors and large
datasets.i address them by exploring the data applying appropriate
```

```
Pandas methods and validating changes.
    data.info()
    print(data.isnull().sum())
    print(data.duplicated().sum())
    print(data.describe())
    print(data["Gender"].unique())
```

8.How can you check data quality?

```
I assess completeness ,consistency ,validity,accuracy and uniqueness.I
check missing values,duplicates,datatypes,formats,unusual values and
business rules then validate the cleaned dataset.
    data.info()
    print(data.isnull().sum())
    print(data.duplicated().sum())
    print(data.describe())
    print(data["Gender"].unique())
```